

Introduction to Open Source Software (009957-001) - Final Presentation Report

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1. Introduction

SmartRemind is a Reflex-based Assignment Reminder AI Agent designed to help students manage academic deadlines more effectively. The system tracks assignments, calculates urgency levels based on due dates, generates contextual reminders, and provides a dashboard for monitoring upcoming and overdue tasks.

The goal of SmartRemind is to reduce missed deadlines and improve student time management through automated deadline awareness and assignment prioritization.

2. Tools and Libraries

Tool / Library	Purpose
Python	Main programming language
Reflex	Builds the web application's user interface
Pydantic	Validates and structures data models
JSON	Stores configuration or application data, depending on how your project uses it.
datetime	Built-in Python module for handling dates and times
typing	Built-in Python module for type hints such as List.

3. Code explanation

```
import reflex as rx
from datetime import date, datetime
from typing import List
from pydantic import BaseModel

#data
class Assignment(BaseModel):
    id: int
    name: str
    subject: str
    due_date: str
    days_remaining: int = 0
    priority: str = "upcoming"
    reminder: str = ""
    done: bool = False
```

The code defines an Assignment data model using Pydantic's BaseModel to structure and validate assignment information in the application. It includes fields such as id, name, subject, and due_date to store basic assignment details. Additional attributes like days_remaining, priority, reminder, and done are used to track the status, urgency, optional reminders, and completion state of each assignment. Default values are assigned to some fields to ensure the model works even when certain information is not provided.

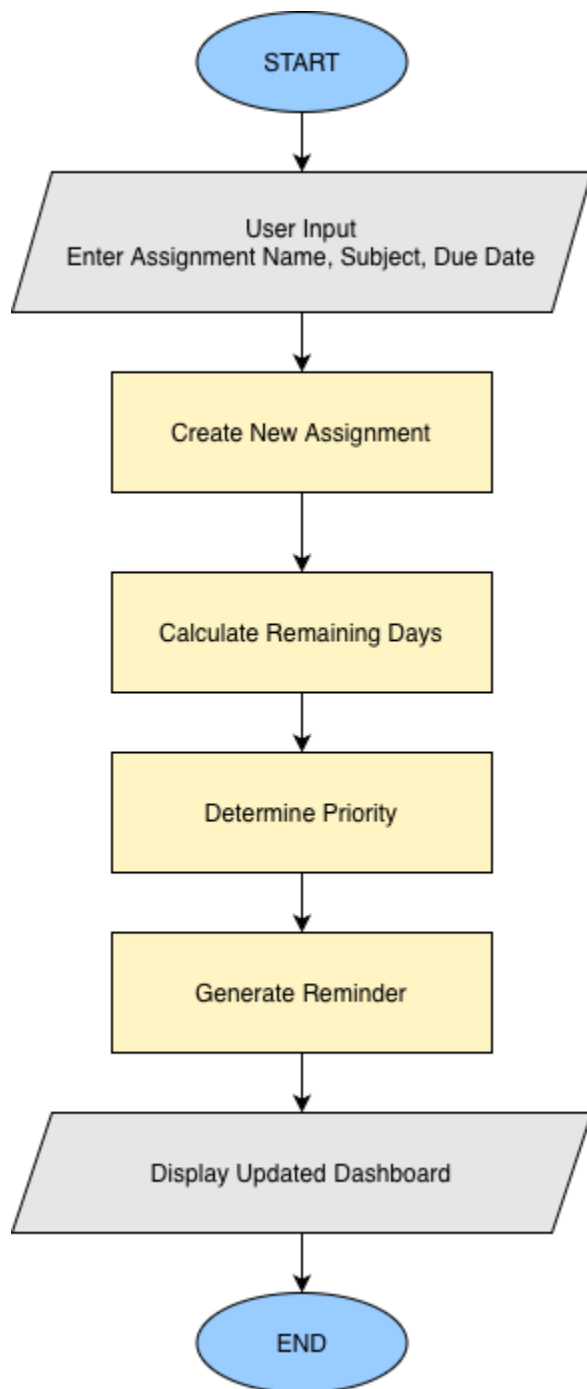
```

5 #calculate date
6 def calc_days(due_date_str: str) -> int:
7     try:
8         due = datetime.strptime(due_date_str, "%Y-%m-%d").date()
9         return (due - date.today()).days
10    except Exception:
11        return 999
12
13
14 #prio
15 def calc_priority(days: int) -> str:
16     if days < 0: return "overdue"
17     if days == 0: return "due_today"
18     if days == 1: return "due_tomorrow"
19     return "upcoming"
20
21
22 #generate reminder
23 def generate_reminder(name: str, subject: str, days: int, priority: str) -> str:
24     if priority == "overdue":
25         return f"🚨 OVERDUE by {abs(days)} day(s)! Submit '{name}' immediately and contact your lecturer."
26     if priority == "due_today":
27         return f"📅 '{name}' ({subject}) is due TODAY! Stop everything and finish it now."
28     if priority == "due_tomorrow":
29         return f"📅 '{name}' ({subject}) is due TOMORROW. Finish it tonight – no more delays!"
30     if days <= 3:
31         return f"🟡 '{name}' ({subject}) is due in {days} days. Start now to avoid last-minute panic."
32     if days <= 7:
33         return f"🟡 '{name}' ({subject}) is due in {days} days. Make a plan and start drafting."
34     return f"🟢 '{name}' ({subject}) is due in {days} days. You have time."
35

```

This part of the code handles deadline calculation, priority assignment, and reminder generation for each assignment. The `calc_days` function calculates how many days are left until the due date by converting the input string into a date and subtracting it from the current date; if an error occurs, it returns a large value (999) to indicate an invalid date. The `calc_priority` function determines the urgency level based on the number of remaining days, categorizing tasks as overdue, due today, due tomorrow, or upcoming. The `generate_reminder` function then creates a user-friendly reminder message with different warning levels depending on the priority, helping users understand how urgent each assignment is and encouraging timely completion.

4. Flowchart



5. Dashboard

SmartRemind

Assignment Reminder

Dashboard & Analytics

3

Total Assignments

0

Overdue

2

Upcoming

New Assignment

Assignment Name

e.g. Lab Report

Subject

e.g. Biology

Due Date

06/14/2026

+ Add Assignment

Filter by subject: All Sort by: due_date

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

AI Agent Report

Artificial Intelligence · 2026-06-14

Due Today

Today!

Data Structures Quiz

Computer Science · 2026-06-16

Upcoming

2 days left

Network Security Essay

Cybersecurity · 2026-06-21

Upcoming

7 days left

Reminders

Generate Reminders

6. Input Example

Adding new assignment

The screenshot shows a 'New Assignment' form with the following fields and values:

- Assignment Name:** Week 7 Task
- Subject:** Open Source
- Due Date:** 06/18/2026

A date picker is open, showing the month of June 2026. The date 18 is highlighted in green. The date picker also shows the days of the week (Su, Mo, Tu, We, Th, Fr, Sa) and the days of the month (1-31).

Below the form, there is a blue button labeled '+ Add Assignment' and a 'Sort by: due_date' dropdown menu.

New assignment displayed in the To-Do List and Reminders

 To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

 **AI Agent Report**

📁 Artificial Intelligence · 📅 2026-06-14

Due Today

Today!

✓



 **Data Structures Quiz**

📁 Computer Science · 📅 2026-06-16

Upcoming

2 days left

✓



 **Week 7 Task**

📁 Open Source · 📅 2026-06-18

Upcoming

4 days left

✓



 **Network Security Essay**

📁 Cybersecurity · 📅 2026-06-21

Upcoming

7 days left

✓



 Reminders

 Generate Reminders

 'AI Agent Report' (Artificial Intelligence) is due TODAY! Stop everything and finish it now.

 'Data Structures Quiz' (Computer Science) is due in 2 days. Start now to avoid last-minute panic.

 'Week 7 Task' (Open Source) is due in 4 days. Make a plan and start drafting.

 'Network Security Essay' (Cybersecurity) is due in 7 days. Make a plan and start drafting.

New subject inserted into the subject list

Filter by subject

due_date

✓ All

Artificial Intelligence

Computer Science

Cybersecurity

Open Source

To-Do List

Agent automatically classifies tasks (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

AI Agent Report

Artificial Intelligence · 2026-06-14

Due Today

Today!

✓

Data Structures Quiz

Computer Science · 2026-06-16

Upcoming

2 days left

✓

Week 7 Task

Open Source · 2026-06-18

Upcoming

4 days left

✓

Network Security Essay

Cybersecurity · 2026-06-21

Upcoming

7 days left

✓

Mark as Done Feature - Assignments marked as done will appear on the Completed Assignments list.

To-Do List

Agent automatically calculates days remaining (Layer 2), classifies urgency (Layer 3), and generates reminders (Layer 4).

Week 7 Task

Open Source · 2026-06-18

Upcoming ✓
4 days left

Network Security Essay

Cybersecurity · 2026-06-21

Upcoming ✓
7 days left

Reminders

Generate Reminders

'Week 7 Task' (Open Source) is due in 4 days. Make a plan and start drafting.

'Network Security Essay' (Cybersecurity) is due in 7 days. Make a plan and start drafting.

Completed Assignments

AI Agent Report

Artificial Intelligence

Data Structures Quiz

Computer Science

Dashboard updated

Dashboard & Analytics

4

Total Assignments

0

Overdue

2

Upcoming

7. Importance of [README.md](#) and requirements.txt in open source project

In an open-source project, both README.md and requirements.txt are important because they help other people understand, install, and use the project easily.

The [README.md](#) file is the main documentation of the project. It explains what the project does, how to set it up, how to run it, and sometimes how to contribute. It acts as the first guide for anyone visiting the repository, making the project more understandable and accessible, especially for new users or developers.

The requirements.txt file lists all the Python libraries needed for the project to run correctly. It ensures that anyone who downloads the project can install the exact dependencies using a single command. This helps avoid compatibility issues and makes sure the project works the same way across different environments.